

Machine Learning MCQs

1. What is Machine Learning (ML)?

- A. The autonomous acquisition of knowledge through the use of manual programs
- B. The selective acquisition of knowledge through the use of computer programs
- C. The selective acquisition of knowledge through the use of manual programs
- D. The autonomous acquisition of knowledge through the use of computer programs

Correct option is D

2. Father of Machine Learning (ML)

- A. Geoffrey Chaucer
- B. Geoffrey Hill
- C. Geoffrey Everest Hinton
- D. None of the above

Correct option is C

3. Which is FALSE regarding regression?

- A. It may be used for interpretation
- B. It is used for prediction
- C. It discovers causal relationships
- D. It relates inputs to outputs

Correct option is C

4. Choose the correct option regarding machine learning (ML) and artificial intelligence (AI)

- A. ML is a set of techniques that turns a dataset into a software
- B. AI is a software that can emulate the human mind
- C. ML is an alternate way of programming intelligent machines
- D. All of the above

Correct option is D

5. Which of the factors affect the performance of the learner system does not include?

- A. Good data structures
- B. Representation scheme used
- C. Training scenario
- D. Type of feedback

Correct option is A

6. In general, to have a well-defined learning problem, we must identify which of the following

- A. The class of tasks
- B. The measure of performance to be improved
- C. The source of experience
- D. All of the above

Correct option is D

7. Successful applications of ML

- A. Learning to recognize spoken words
- B. Learning to drive an autonomous vehicle
- C. Learning to classify new astronomical structures
- D. Learning to play world-class backgammon
- E. All of the above

Correct option is E

8. Which of the following does not include different learning methods

- A. Analogy
- B. Introduction
- C. Memorization
- D. Deduction

Correct option is B

9. In language understanding, the levels of knowledge that does not include?

- A. Empirical
- B. Logical
- C. Phonological
- D. Syntactic

Correct option is A

10. Designing a machine learning approach involves:-

- A. Choosing the type of training experience
- B. Choosing the target function to be learned
- C. Choosing a representation for the target function
- D. Choosing a function approximation algorithm
- E. All of the above

Correct option is E

11. Concept learning infers a _____ valued function from training examples of its input and output.

- A. Decimal
- B. Hexadecimal
- C. Boolean
- D. All of the above

Correct option is C

12. Which of the following is not a supervised learning?

- A. Naive Bayesian
- B. PCA
- C. Linear Regression
- D. Decision Tree Answer

Correct option is B

13. What is Machine Learning?

- Artificial Intelligence
 - Deep Learning
 - Data Statistics
- A. Only (i)
 - B. (i) and (ii)
 - C. All
 - D. None

Correct option is B

14. What kind of learning algorithm for “Facial identities or facial expressions”?

- A. Prediction
- B. Recognition Patterns
- C. Generating Patterns
- D. Recognizing Anomalies Answer

Correct option is B

15. Which of the following is not type of learning?

- A. Unsupervised Learning
- B. Supervised Learning
- C. Semi-supervised Learning
- D. Reinforcement Learning

Correct option is C

16. Real-Time decisions, Game AI, Learning Tasks, Skill Acquisition, and Robot Navigation are applications of which of the following

- A. Supervised Learning: Classification
- B. Reinforcement Learning
- C. Unsupervised Learning: Clustering
- D. Unsupervised Learning: Regression

Correct option is B

17. Targeted marketing, Recommended Systems, and Customer Segmentation are applications in which of the following

- A. Supervised Learning: Classification

- B. Unsupervised Learning: Clustering
- C. Unsupervised Learning: Regression
- D. Reinforcement Learning

Correct option is B

18. Fraud Detection, Image Classification, Diagnostic, and Customer Retention are applications in which of the following

- A. Unsupervised Learning: Regression
- B. Supervised Learning: Classification
- C. Unsupervised Learning: Clustering
- D. Reinforcement Learning

Correct option is B

19. Which of the following is not a function of symbolic in the various function representation of Machine Learning?

- A. Rules in propositional Logic
- B. Hidden-Markov Models (HMM)
- C. Rules in first-order predicate logic
- D. Decision Trees

Correct option is B

20. Which of the following is not a numerical function in the various function representation of Machine Learning?

- A. Neural Network
- B. Support Vector Machines
- C. Case-based
- D. Linear Regression

Correct option is C

21. FIND-S Algorithm starts from the most specific hypothesis and generalizes it by considering only_

- A. Negative
- B. Positive
- C. Negative or Positive
- D. None of the above

Correct option is B

22. FIND-S algorithm ignores_

- A. Negative
- B. Positive
- C. Both
- D. None of the above

Correct option is A

23.The Candidate-Elimination Algorithm represents the_.

- A. Solution Space
- B. Version Space
- C. Elimination Space
- D. All of the above

Correct option is B

24.Inductive learning is based on the knowledge that if something happens a lot it is likely to be generally

- A. True
- B. False Answer

Correct option is A

25.Inductive learning takes examples and generalizes rather than starting with _____

- A. Inductive
- B. Existing
- C. Deductive
- D. None of these

Correct option is B

26.A drawback of the FIND-S is that it assumes the consistency within the training set

- A. True
- B. False

Correct option is A

27.What strategies can help reduce overfitting in decision trees?

- Enforce a maximum depth for the tree
- Enforce a minimum number of samples in leaf nodes
- Pruning
- Make sure each leaf node is one pure class

- A. All
- B. (i), (ii) and (iii)
- C. (i), (iii), (iv)
- D. None

Correct option is B

28.Which of the following is a widely used and effective machine learning algorithm based on the idea of bagging?

- A. Decision Tree
- B. Random Forest
- C. Regression
- D. Classification

Correct option is B

29.To find the minimum or the maximum of a function, we set the gradient to zero because which of the following

- A. Depends on the type of problem
- B. The value of the gradient at extrema of a function is always zero
- C. Both (A) and (B)
- D. None of these

Correct option is B

30.Which of the following is a disadvantage of decision trees?

- A. Decision trees are prone to be overfit
- B. Decision trees are robust to outliers
- C. Factor analysis
- D. None of the above

Correct option is A

31.What is perceptron?

- A. A single layer feed-forward neural network with pre-processing
- B. A neural network that contains feedback
- C. A double layer auto-associative neural network
- D. An auto-associative neural network

Correct option is A

32.Which of the following is true for neural networks?

- The training time depends on the size of the
 - Neural networks can be simulated on a conventional
 - Artificial neurons are identical in operation to biological
- A. All
 - B. Only (ii)
 - C. (i) and (ii)
 - D. None

Correct option is C

33.What are the advantages of neural networks over conventional computers?

- They have the ability to learn by
 - They are more fault
 - They are more suited for real time operation due to their high „computational“
- A. (i) and (ii)
 - B. (i) and (iii)
 - C. Only (i)
 - D. All
 - E. None

Correct option is D

34.What is Neuro software?

- A. It is software used by Neurosurgeon
- B. Designed to aid experts in real world
- C. It is powerful and easy neural network
- D. A software used to analyze neurons

Correct option is C

35. Which is true for neural networks?

- A. Each node computes its weighted input
- B. Node could be in excited state or non-excited state
- C. It has set of nodes and connections
- D. All of the above

Correct option is D

36. What is the objective of backpropagation algorithm?

- A. To develop learning algorithm for multilayer feedforward neural network, so that network can be trained to capture the mapping implicitly
- B. To develop learning algorithm for multilayer feedforward neural network
- C. To develop learning algorithm for single layer feedforward neural network
- D. All of the above

Correct option is A

37. Which of the following is true?

Single layer associative neural networks do not have the ability to:-

- Perform pattern recognition
 - Find the parity of a picture
 - Determine whether two or more shapes in a picture are connected or not
- A. (ii) and (iii)
 - B. Only (ii)
 - C. All
 - D. None

Correct option is A

38. The backpropagation law is also known as generalized delta rule

- A. True
- B. False

Correct option is A

38. Which of the following is true?

- On average, neural networks have higher computational rates than conventional computers.
 - Neural networks learn by
 - Neural networks mimic the way the human brain
- A. All
 - B. (ii) and (iii)
 - C. (i), (ii) and (iii)

D. None

Correct option is A

39. What is true regarding backpropagation rule?

- A. Error in output is propagated backwards only to determine weight updates
- B. There is no feedback of signal at any stage
- C. It is also called generalized delta rule
- D. All of the above

Correct option is D

40. There is feedback in final stage of backpropagation

- A. True
- B. False

Correct option is B

41. An auto-associative network is

- A. A neural network that has only one loop
- B. A neural network that contains feedback
- C. A single layer feed-forward neural network with pre-processing
- D. A neural network that contains no loops

Correct option is B

42. A 3-input neuron has weights 1, 4 and 3. The transfer function is linear with the constant of proportionality being equal to 3. The inputs are 4, 8 and 5 respectively. What will be the output?

- A. 139
- B. 153
- C. 162
- D. 160

Correct option is B

43. What of the following is true regarding backpropagation rule?

- A. Hidden layers output is not all important, they are only meant for supporting input and output layers
- B. Actual output is determined by computing the outputs of units for each hidden layer
- C. It is a feedback neural network
- D. None of the above

Correct option is B

44. What is back propagation?

- A. It is another name given to the curvy function in the perceptron
- B. It is the transmission of error back through the network to allow weights to be adjusted so that the network can learn
- C. It is another name given to the curvy function in the perceptron

D. None of the above

Correct option is B

45. The general limitations of back propagation rule is/are

- A. Scaling
- B. Slow convergence
- C. Local minima problem
- D. All of the above

Correct option is D

46. What is the meaning of generalized in statement “backpropagation is a generalized delta rule” ?

- A. Because delta is applied to only input and output layers, thus making it more simple and generalized
- B. It has no significance
- C. Because delta rule can be extended to hidden layer units
- D. None of the above

Correct option is C

47. Neural Networks are complex _____ functions with many parameters

- A. Linear
- B. Non linear
- C. Discrete
- D. Exponential

Correct option is A

48. The general tasks that are performed with backpropagation algorithm

- A. Pattern mapping
- B. Prediction
- C. Function approximation
- D. All of the above

Correct option is D

49. Backpropagation learning is based on the gradient descent along error surface.

- A. True
- B. False

Correct option is A

50. In backpropagation rule, how to stop the learning process?

- A. No heuristic criteria exist
- B. On basis of average gradient value
- C. There is convergence involved
- D. None of these

Correct option is B

51.Applications of NN (Neural Network)

- A. Risk management
- B. Data validation
- C. Sales forecasting
- D. All of the above

Correct option is D

52.The network that involves backward links from output to the input and hidden layers is known as

- A. Recurrent neural network
- B. Self organizing maps
- C. Perceptrons
- D. Single layered perceptron

Correct option is A

53.Decision Tree is a display of an Algorithm?

- A. True
- B. False

Correct option is A

54.Which of the following is/are the decision tree nodes?

- A. End Nodes
- B. Decision Nodes
- C. Chance Nodes
- D. All of the above

Correct option is D

55.End Nodes are represented by which of the following

- A. Solar street light
- B. Triangles
- C. Circles
- D. Squares

Correct option is B

56.Decision Nodes are represented by which of the following

- A. Solar street light
- B. Triangles
- C. Circles
- D. Squares

Correct option is D

57.Chance Nodes are represented by which of the following

- A. Solar street light
- B. Triangles
- C. Circles
- D. Squares

Correct option is C

58. Advantage of Decision Trees

- A. Possible Scenarios can be added
- B. Use a white box model, if given result is provided by a model
- C. Worst, best and expected values can be determined for different scenarios
- D. All of the above

Correct option is D

59. _____ terms are required for building a bayes model.

- A. 1
- B. 2
- C. 3
- D. 4

Correct option is C

60. Which of the following is the consequence between a node and its predecessors while creating bayesian network?

- A. Conditionally independent
- B. Functionally dependent
- C. Both Conditionally dependant & Dependant
- D. Dependent

Correct option is A

61. Why it is needed to make probabilistic systems feasible in the world?

- A. Feasibility
- B. Reliability
- C. Crucial robustness
- D. None of the above

Correct option is C

62. Bayes rule can be used for:-

- A. Solving queries
- B. Increasing complexity
- C. Answering probabilistic query
- D. Decreasing complexity

Correct option is C

63. _____ provides way and means of weighing up the desirability of goals and the likelihood of achieving

- A. Utility theory
- B. Decision theory
- C. Bayesian networks
- D. Probability theory

Correct option is A

64. Which of the following is provided by the Bayesian Network?

- A. Complete description of the problem
- B. Partial description of the domain
- C. Complete description of the domain
- D. All of the above

Correct option is C

65. Probability provides a way of summarizing the _____ that comes from our laziness and

- A. Belief
- B. Uncertainty
- C. Joint probability distributions
- D. Randomness

Correct option is B

66. The entries in the full joint probability distribution can be calculated as

- A. Using variables
- B. Both Using variables & information
- C. Using information
- D. All of the above

Correct option is C

67. Causal chain (For example, Smoking causes cancer) gives rise to:-

- A. Conditionally Independence
- B. Conditionally Dependence
- C. Both
- D. None of the above

Correct option is A

68. The Bayesian network can be used to answer any query by using:-

- A. Full distribution
- B. Joint distribution
- C. Partial distribution
- D. All of the above

Correct option is B

69. Bayesian networks allow compact specification of:-

- A. Joint probability distributions
- B. Belief

- C. Propositional logic statements
- D. All of the above

Correct option is A

70. The compactness of the bayesian network can be described by

- A. Fully structured
- B. Locally structured
- C. Partially structured
- D. All of the above

Correct option is B

71. The Expectation-Maximization Algorithm has been used to identify conserved domains in unaligned proteins only. State True or False.

- A. True
- B. False

Correct option is B

72. Which of the following is correct about the Naive Bayes?

- A. Assumes that all the features in a dataset are independent
- B. Assumes that all the features in a dataset are equally important
- C. Both
- D. All of the above

Correct option is C

73. Which of the following is false regarding EM Algorithm?

- A. The alignment provides an estimate of the base or amino acid composition of each column in the site
- B. The column-by-column composition of the site already available is used to estimate the probability of finding the site at any position in each of the sequences
- C. The row-by-column composition of the site already available is used to estimate the probability
- D. None of the above

Correct option is C

74. Naïve Bayes Algorithm is a learning algorithm.

- A. Supervised
- B. Reinforcement
- C. Unsupervised
- D. None of these

Correct option is A

75. EM algorithm includes two repeated steps, here the step 2 is_____.

- A. The normalization
- B. The maximization step

- C. The minimization step
- D. None of the above

Correct option is C

76.Examples of Naïve Bayes Algorithm is/are

- A. Spam filtration
- B. Sentimental analysis
- C. Classifying articles
- D. All of the above

Correct option is D

77.In the intermediate steps of “EM Algorithm”, the number of each base in each column is determined and then converted to

- A. True
- B. False

Correct option is A

78.Naïve Bayes algorithm is based on and used for solving classification problems.

- A. Bayes Theorem
- B. Candidate elimination algorithm
- C. EM algorithm
- D. None of the above

Correct option is A

79.Types of Naïve Bayes Model:

- A. Gaussian
- B. Multinomial
- C. Bernoulli
- D. All of the above

Correct option is D

80.Disadvantages of Naïve Bayes Classifier:

- A. Naive Bayes assumes that all features are independent or unrelated, so it cannot learn the relationship between
- B. It performs well in Multi-class predictions as compared to the other
- C. Naïve Bayes is one of the fast and easy ML algorithms to predict a class of
- D. It is the most popular choice for text classification problems.

Correct option is A

81.The benefit of Naïve Bayes:-

- A. Naïve Bayes is one of the fast and easy ML algorithms to predict a class of
- B. It is the most popular choice for text classification problems.
- C. It can be used for Binary as well as Multi-class
- D. All of the above

Correct option is D

82. In which of the following types of sampling the information is carried out under the opinion of an expert?

- A. Convenience sampling
- B. Judgement sampling
- C. Quota sampling
- D. Purposive sampling

Correct option is B

83. Full form of MDL?

- A. Minimum Description Length
- B. Maximum Description Length
- C. Minimum Domain Length
- D. None of these

Correct option is A

84. For the analysis of ML algorithms, we need

- A. Computational learning theory
- B. Statistical learning theory
- C. Both A & B
- D. None of these

Correct option is C

85. PAC stand for

- A. Probably Approximate Correct
- B. Probably Approx Correct
- C. Probably Approximate Computation
- D. Probably Approx Computation

Correct option is A

86. _____ hypothesis h with respect to target concept c and distribution D , is the probability that h will misclassify an instance drawn at random according to D .

- A. True Error
- B. Type 1 Error
- C. Type 2 Error
- D. None of these

Correct option is A

87. Statement: True error defined over entire instance space, not just training data

- A. True
- B. False

Correct option is A

88.What are the area CLT comprised of?

- A. Sample Complexity
- B. Computational Complexity
- C. Mistake Bound
- D. All of these

Correct option is D

88.What area of CLT tells “How many examples we need to find a good hypothesis ?”?

- A. Sample Complexity
- B. Computational Complexity
- C. Mistake Bound
- D. None of these

Correct option is A

89.What area of CLT tells “How much computational power we need to find a good hypothesis ?”?

- A. Sample Complexity
- B. Computational Complexity
- C. Mistake Bound
- D. None of these

Correct option is B

90.What area of CLT tells “How many mistakes we will make before finding a good hypothesis ?”?

- A. Sample Complexity
- B. Computational Complexity
- C. Mistake Bound
- D. None of these

Correct option is C

91.(For question no. 9 and 10) Can we say that concept described by conjunctions of Boolean literals are PAC learnable?

- A. Yes
- B. No

Correct option is A

92.How large is the hypothesis space when we have n Boolean attributes?

- A. $|H| = 3^n$
- B. $|H| = 2^n$
- C. $|H| = 1^n$
- D. $|H| = 4^n$

Correct option is A

93. The VC dimension of hypothesis space H_1 is larger than the VC dimension of hypothesis space H_2 . Which of the following can be inferred from this?

- A. The number of examples required for learning a hypothesis in H_1 is larger than the number of examples required for H_2
- B. The number of examples required for learning a hypothesis in H_1 is smaller than the number of examples required for H_2
- C. No relation to number of samples required for PAC learning.

Correct option is A

94. For a particular learning task, if the requirement of error parameter changes from 0.1 to 0.01. How many more samples will be required for PAC learning?

- A. Same
- B. 2 times
- C. 1000 times
- D. 10 times

Correct option is D

95. Computational complexity of classes of learning problems depends on which of the following?

- A. The size or complexity of the hypothesis space considered by learner
- B. The accuracy to which the target concept must be approximated
- C. The probability that the learner will output a successful hypothesis
- D. All of these

Correct option is D

96. The instance-based learner is a _____

- A. Lazy-learner
- B. Eager learner
- C. Can't say

Correct option is A

97. When to consider nearest neighbour algorithms?

- A. Instance map to point in k^n
- B. Not more than 20 attributes per instance
- C. Lots of training data
- D. None of these
- E. A, B & C

Correct option is E

98. What are the advantages of Nearest neighbour algo?

- A. Training is very fast
- B. Can learn complex target functions
- C. Don't lose information

D. All of these

Correct option is D

99. What are the difficulties with k-nearest neighbour algo?

- A. Calculate the distance of the test case from all training cases
- B. Curse of dimensionality
- C. Both A & B
- D. None of these

Correct option is C

100. What if the target function is real valued in kNN algo?

- A. Calculate the mean of the k nearest neighbours
- B. Calculate the SD of the k nearest neighbour
- C. None of these

Correct option is A

101. What is/are true about Distance-weighted KNN?

- A. The weight of the neighbour is considered
- B. The distance of the neighbour is considered
- C. Both A & B
- D. None of these

Correct option is C

102. What is/are advantage(s) of Distance-weighted k-NN over k-NN?

- A. Robust to noisy training data
- B. Quite effective when a sufficient large set of training data is provided
- C. Both A & B
- D. None of these

Correct option is C

103. What is/are advantage(s) of Locally Weighted Regression?

- A. Pointwise approximation of complex target function
- B. Earlier data has no influence on the new ones
- C. Both A & B
- D. None of these

Correct option is C

104. The quality of the result depends on (LWR)

- A. Choice of the function
- B. Choice of the kernel function K
- C. Choice of the hypothesis space H
- D. All of these

Correct option is D

105.How many types of layer in radial basis function neural networks?

- A. 3
- B. 2
- C. 1
- D. 4

Correct option is A, Input layer, Hidden layer, and Output layer

106.The neurons in the hidden layer contains Gaussian transfer function whose output are_____to the distance from the centre of the neuron.

- A. Directly
- B. Inversely
- C. equal
- D. None of these

Correct option is B

107.PNN/GRNN networks have one neuron for each point in the training file, While RBF network have a variable number of neurons that is usually

- A. less than the number of training
- B. greater than the number of training points
- C. equal to the number of training points
- D. None of these

Correct option is A

108.Which network is more accurate when the size of training set between small to medium?

- A. PNN/GRNN
- B. RBF
- C. K-means clustering
- D. None of these

Correct option is A

109.What is/are true about RBF network?

- A. A kind of supervised learning
- B. Design of NN as curve fitting problem
- C. Use of multidimensional surface to interpolate the test data
- D. All of these

Correct option is D

110.Application of CBR

- A. Design
- B. Planning
- C. Diagnosis
- D. All of these

Correct option is A

111.What is/are advantages of CBR?

- A. A local approx. is found for each test case
- B. Knowledge is in a form understandable to human
- C. Fast to train
- D. All of these

Correct option is D

112 In k-NN algorithm, given a set of training examples and the value of $k < \text{size of training set } (n)$, the algorithm predicts the class of a test example to be the. What is/are advantages of CBR?

- A. Least frequent class among the classes of k closest training
- B. Most frequent class among the classes of k closest training
- C. Class of the closest
- D. Most frequent class among the classes of the k farthest training examples.

Correct option is B

113.Which of the following statements is true about PCA?

- We must standardize the data before applying
 - We should select the principal components which explain the highest variance
 - We should select the principal components which explain the lowest variance
 - We can use PCA for visualizing the data in lower dimensions
- A. (i), (ii) and (iv).
 - B. (ii) and (iv)
 - C. (iii) and (iv)
 - D. (i) and (iii)

Correct option is A

114.Genetic algorithm is a

- A. Search technique used in computing to find true or approximate solution to optimization and search problem
- B. Sorting technique used in computing to find true or approximate solution to optimization and sort problem
- C. Both A & B
- D. None of these

Correct option is A

115.GA techniques are inspired by_

- A. Evolutionary
- B. Cytology
- C. Anatomy
- D. Ecology

Correct option is A

116.When would the genetic algorithm terminate?

- A. Maximum number of generations has been produced

- B. Satisfactory fitness level has been reached for the
- C. Both A & B
- D. None of these

Correct option is C

117.The algorithm operates by iteratively updating a pool of hypotheses, called the

- A. Population
- B. Fitness
- C. None of these

Correct option is A

118.What is the correct representation of GA?

- A. GA(Fitness, Fitness_threshold, p)
- B. GA(Fitness, Fitness_threshold, p, r)
- C. GA(Fitness, Fitness_threshold, p, r, m)
- D. GA(Fitness, Fitness_threshold)

Correct option is C

119.Genetic operators includes

- A. Crossover
- B. Mutation
- C. Both A & B
- D. None of these

Correct option is C

120.Produces two new offspring from two parent string by copying selected bits from each parent is called

- A. Mutation
- B. Inheritance
- C. Crossover
- D. None of these

Correct option is C

121.Each schema the set of bit strings containing the indicated as

- A. 0s, 1s
- B. only 0s
- C. only 1s
- D. 0s, 1s, *s

Correct option is D

122.0*10 represents the set of bit strings that includes exactly (A) 0010, 0110

- A. 0010, 0010
- B. 0100, 0110
- C. 0100, 0010

Correct option is A

123. Correct (h) is the percent of all training examples correctly classified by hypothesis then Fitness function is equal to

- A. $\text{Fitness (h)} = (\text{correct (h)})^2$
- B. $\text{Fitness (h)} = (\text{correct (h)})^3$
- C. $\text{Fitness (h)} = (\text{correct (h)})$
- D. $\text{Fitness (h)} = (\text{correct (h)})^4$

Correct option is A

124. Statement: Genetic Programming individuals in the evolving population are computer programs rather than bit

- A. True
- B. False

Correct option is A

125. _____ evolution over many generations was directly influenced by the experiences of individual organisms during their lifetime

- A. Baldwin
- B. Lamarckian
- C. Bayes
- D. None of these

Correct option is B

126. Search through the hypothesis space cannot be characterized. Why?

- A. Hypotheses are created by crossover and mutation operators that allow radical changes between successive generations
- B. Hypotheses are not created by crossover and mutation
- C. None of these

Correct option is A

127. ILP stand for

- A. Inductive Logical programming
- B. Inductive Logic Programming
- C. Inductive Logical Program
- D. Inductive Logic Program

Correct option is B

128. What is/are the requirement for the Learn-One-Rule method?

- A. Input, accepts a set of +ve and -ve training examples.
- B. Output, delivers a single rule that covers many +ve examples and few -ve.
- C. Output rule has a high accuracy but not necessarily a high
- D. A & B
- E. A, B & C

Correct option is E

129. _____ is any predicate (or its negation) applied to any set of terms.

- A. Literal
- B. Null
- C. Clause
- D. None of these

Correct option is A

130. Ground literal is a literal that

- A. Contains only variables
- B. does not contains any functions
- C. does not contains any variables
- D. Contains only functions Answer

Correct option is C

131. _____ emphasizes learning feedback that evaluates the learner's performance without providing standards of correctness in the form of behavioural

- A. Reinforcement learning
- B. Supervised Learning
- C. None of these

Correct option is A

132. Features of Reinforcement learning

- A. Set of problem rather than set of techniques
- B. RL is training by reward and
- C. RL is learning from trial and error with the
- D. All of these

Correct option is D

133. Which type of feedback used by RL?

- A. Purely Instructive feedback
- B. Purely Evaluative feedback
- C. Both A & B
- D. None of these

Correct option is B

134. What is/are the problem solving methods for RL?

- A. Dynamic programming
- B. Monte Carlo Methods
- C. Temporal-difference learning
- D. All of these

Correct option is D

135.The FIND-S Algorithm

- A. Starts with starts from the most specific hypothesis Answer
- B. It considers negative examples
- C. It considers both negative and positive
- D. None of these Correct

136. The hypothesis space has a general-to-specific ordering of hypotheses, and the search can be efficiently organized by taking advantage of a naturally occurring structure over the hypothesis space

- A. TRUE
- B. FALSE

Correct option is A

137. The Version space is:

- A. The subset of all hypotheses is called the version space with respect to the hypothesis space H and the training examples D , because it contains all plausible versions of the target
- B. The version space consists of only specific
- C. None of these
- D. All of the Above

Correct option is A

138.The Candidate-Elimination Algorithm

- A. The key idea in the Candidate-Elimination algorithm is to output a description of the set of all hypotheses consistent with the training
- B. Candidate-Elimination algorithm computes the description of this set without explicitly enumerating all of its
- C. This is accomplished by using the more-general-than partial ordering and maintaining a compact representation of the set of consistent
- D. All of these

Correct option is D

139.Concept learning is basically acquiring the definition of a general category from given sample positive and negative training examples of the

- A. TRUE
- B. FALSE

Correct option is A

140.The hypothesis h_1 is more-general-than hypothesis h_2 ($h_1 > h_2$) if and only if $h_1 \geq h_2$ is true and $h_2 \geq h_1$ is false. We also say h_2 is more-specific-than h_1

- A. The statement is true
- B. The statement is false
- C. We cannot
- D. None of these

Correct option is A

141.The List-Then-Eliminate Algorithm

- A. The List-Then-Eliminate algorithm initializes the version space to contain all hypotheses in H, then eliminates any hypothesis found inconsistent with any training
- B. The List-Then-Eliminate algorithm not initializes to the version
- C. None of these Answer

Correct option is A

142.What will take place as the agent observes its interactions with the world?

- A. Learning
- B. Hearing
- C. Perceiving
- D. Speech

Correct option is A

143.Which modifies the performance element so that it makes better decision?Performance element

- A. Performance element
- B. Changing element
- C. Learning element
- D. None of the mentioned

Correct option is C

144.Any hypothesis found to approximate the target function well over a sufficiently large set of training examples will also approximate the target function well over other unobserved example is called:

- A. Inductive Learning Hypothesis
- B. Null Hypothesis
- C. Actual Hypothesis
- D. None of these

Correct option is A

145.Feature of ANN in which ANN creates its own organization or representation of information it receives during learning time is

- A. Adaptive Learning
- B. Self Organization
- C. What-If Analysis
- D. Supervised Learning

Correct option is B

146.How the decision tree reaches its decision?

- A. Single test
- B. Two test
- C. Sequence of test
- D. No test

Correct option is C

147.Which of the following is a disadvantage of decision trees?

- Factor analysis
- Decision trees are robust to outliers
- Decision trees are prone to be overfit
- None of the above

Correct option is C

148. Tree/Rule based classification algorithms generate which rule to perform the classification.

- A. if-then.
- B. then
- C. do
- D. Answer

Correct option is A

149. What is Gini Index?

- A. It is a type of index structure
- B. It is a measure of purity
- C. None of the options

Correct option is A

150. What is not a RNN in machine learning?

- A. One output to many inputs
- B. Many inputs to a single output
- C. RNNs for nonsequential input
- D. Many inputs to many outputs

Correct option is A

151. Which of the following sentences are correct in reference to Information gain?

- A. It is biased towards multi-valued attributes
- B. ID3 makes use of information gain
- C. The approach used by ID3 is greedy
- D. All of these

Correct option is D

152. A Neural Network can answer

- A. For Loop questions
- B. what-if questions
- C. IF-The-Else Analysis Questions
- D. None of these Answer

Correct option is B

153. Artificial neural network used for

- A. Pattern Recognition
- B. Classification
- C. Clustering
- D. All Answer

Correct option is D

154. Which of the following are the advantage/s of Decision Trees?

- A. Possible Scenarios can be added
- B. Use a white box model, If given result is provided by a model
- C. Worst, best and expected values can be determined for different scenarios
- D. All of the mentioned

Correct option is D

155. What is the mathematical likelihood that something will occur?

- A. Classification
- B. Probability
- C. Naïve Bayes Classifier
- D. None of the other

Correct option is C

- A. What does the Bayesian network provides?
- B. Complete description of the domain
- C. Partial description of the domain
- D. Complete description of the problem
- E. None of the mentioned

Correct option is C

157. Where does the Bayes rule can be used?

- A. Solving queries
- B. Increasing complexity
- C. Decreasing complexity
- D. Answering probabilistic query

Correct option is D

158. How many terms are required for building a Bayes model?

- A. 2
- B. 3
- C. 4
- D. 1

Correct option is B

159. What is needed to make probabilistic systems feasible in the world?

- A. Reliability
- B. Crucial robustness
- C. Feasibility
- D. None of the mentioned

Correct option is B

160. It was shown that the Naive Bayesian method

- A. Can be much more accurate than the optimal Bayesian method
- B. Is always worse off than the optimal Bayesian method
- C. Can be almost optimal only when attributes are independent

D. Can be almost optimal when some attributes are dependent

Correct option is C

161. What is the consequence between a node and its predecessors while creating Bayesian network?

- A. Functionally dependent
- B. Dependant
- C. Conditionally independent
- D. Both Conditionally dependant & Dependant

Correct option is C

162. How the compactness of the Bayesian network can be described?

- A. Locally structured
- B. Fully structured
- C. Partial structure
- D. All of the mentioned

Correct option is A

163. How the entries in the full joint probability distribution can be calculated?

- A. Using variables
- B. Using information
- C. Both Using variables & information
- D. None of the mentioned

Correct option is B

164. How the Bayesian network can be used to answer any query?

- A. Full distribution
- B. Joint distribution
- C. Partial distribution
- D. All of the mentioned

Correct option is B

165. Sample Complexity is

- A. The sample complexity is the number of training-samples that we need to supply to the algorithm, so that the function returned by the algorithm is within an arbitrarily small error of the best possible function, with probability arbitrarily close to 1
- B. How many training examples are needed for learner to converge to a successful hypothesis.
- C. All of these

Correct option is C

166. PAC stands for

- A. Probability Approximately Correct
- B. Probability Applied Correctly
- C. Partition Approximately Correct

Correct option is A

167. Which of the following will be true about k in k -NN in terms of variance

- A. When you increase the k the variance will increase
- B. When you decrease the k the variance will increase
- C. Can't say
- D. None of these

Correct option is B

168. Which of the following option is true about k -NN algorithm?

- A. It can be used for classification
- B. It can be used for regression
- C. It can be used in both classification and regression Answer

Correct option is C

169. In k -NN it is very likely to overfit due to the curse of dimensionality. Which of the following option would you consider to handle such problem? 1). Dimensionality Reduction
2). Feature selection

- A. 1
- B. 2
- C. 1 and 2
- D. None of these

Correct option is C

170. When you find noise in data which of the following option would you consider in k -NN

- A. I will increase the value of k
- B. I will decrease the value of k
- C. Noise can not be dependent on value of k
- D. None of these

Correct option is A

171. Which of the following will be true about k in k -NN in terms of Bias?

- A. When you increase the k the bias will increase
- B. When you decrease the k the bias will increase
- C. Can't say
- D. None of these

Correct option is A

172. What is used to mitigate overfitting in a test set?

- A. Overfitting set
- B. Training set
- C. Validation dataset
- D. Evaluation set

Correct option is C

173. A radial basis function is a

- A. Activation function
- B. Weight

- C. Learning rate
- D. none

Correct option is A

174. Mistake Bound is

- A. How many training examples are needed for learner to converge to a successful hypothesis.
- B. How much computational effort is needed for a learner to converge to a successful hypothesis
- C. How many training examples will the learner misclassify before converging to a successful hypothesis
- D. None of these

Correct option is C

175. All of the following are suitable problems for genetic algorithms EXCEPT

- A. dynamic process control
- B. pattern recognition with complex patterns
- C. simulation of biological models
- D. simple optimization with few variables

Correct option is D

176. Adding more basis functions in a linear model... (Pick the most probably option)

- A. Decreases model bias
- B. Decreases estimation bias
- C. Decreases variance
- D. Doesn't affect bias and variance

Correct option is A

177. Which of these are types of crossover

- A. Single point
- B. Two point
- C. Uniform
- D. All of these

Correct option is D

178. A feature F1 can take certain value: A, B, C, D, E, & F and represents grade of students from a college. Which of the following statement is true in following case?

- A. Feature F1 is an example of nominal
- B. Feature F1 is an example of ordinal
- C. It doesn't belong to any of the above category.

Correct option is B

179. You observe the following while fitting a linear regression to the data: As you increase the amount of training data, the test error decreases and the training error increases. The train error is quite low (almost what you expect it to), while the test error is much higher than the train error. What do you think is the main reason behind this behaviour? Choose the most probable option.

- A. High variance
- B. High model bias
- C. High estimation bias
- D. None of the above Answer

Correct option is C

180. Genetic algorithms are heuristic methods that do not guarantee an optimal solution to a problem

- A. TRUE
- B. FALSE

Correct option is A

181. Which of the following statements about regularization is not correct?

- A. Using too large a value of lambda can cause your hypothesis to underfit the
- B. Using too large a value of lambda can cause your hypothesis to overfit the
- C. Using a very large value of lambda cannot hurt the performance of your hypothesis.
- D. None of the above

Correct option is A

182. Consider the following: (a) Evolution (b) Selection (c) Reproduction (d) Mutation Which of the following are found in genetic algorithms?

- A. All
- B. a, b, c
- C. a, b
- D. b, d

Correct option is A

183. Genetic Algorithm are a part of

- A. Evolutionary Computing
- B. inspired by Darwin's theory about evolution – "survival of the fittest"
- C. are adaptive heuristic search algorithm based on the evolutionary ideas of natural selection and genetics
- D. All of the above

Correct option is D

184. Genetic algorithms belong to the family of methods in the

- A. artificial intelligence area
- B. optimization
- C. complete enumeration family of methods
- D. Non-computer based (human) solutions area

Correct option is A

185. For a two player chess game, the environment encompasses the opponent

- A. True
- B. False

Correct option is A

186. Which among the following is not a necessary feature of a reinforcement learning solution to a learning problem?

- A. exploration versus exploitation dilemma
- B. trial and error approach to learning
- C. learning based on rewards
- D. representation of the problem as a Markov Decision Process

Correct option is D

187. Which of the following sentence is FALSE regarding reinforcement learning

- A. It relates inputs to
- B. It is used for
- C. It may be used for
- D. It discovers causal relationships.

Correct option is D

188. The EM algorithm is guaranteed to never decrease the value of its objective function on any iteration

- A. TRUE
- B. FALSE Answer

Correct option is A

189. Consider the following modification to the tic-tac-toe game: at the end of game, a coin is tossed and the agent wins if a head appears regardless of whatever has happened in the game. Can reinforcement learning be used to learn an optimal policy of playing Tic-Tac-Toe in this case?

- A. Yes
- B. No

Correct option is B

190. Out of the two repeated steps in EM algorithm, the step 2 is _____

- A. the maximization step
- B. the minimization step
- C. the optimization step
- D. the normalization step

Correct option is A

191. Suppose the reinforcement learning player was greedy, that is, it always played the move that brought it to the position that it rated the best. Might it learn to play better, or worse, than a non greedy player?

- A. Worse
- B. Better

Correct option is B

192. A chess agent trained by using Reinforcement Learning can be trained by playing against a copy of the same

- A. True

B. False

Correct option is A

193.The EM iteration alternates between performing an expectation (E) step, which creates a function for the expectation of the log-likelihood evaluated using the current estimate for the parameters, and a maximization (M) step, which computes parameters maximizing the expected log-likelihood found on the E

A. TRUE

B. FALSE

Correct option is A

194.Expectation–maximization (EM) algorithm is an

A. Iterative

B. Incremental

C. None

Correct option is A

195.Feature need to be identified by using Well Posed Learning Problem:

A. Class of tasks

B. Performance measure

C. Training experience

D. All of these

Correct option is D

196.A computer program that learns to play checkers might improve its performance as:

A. Measured by its ability to win at the class of tasks involving playing checkers

B. Experience obtained by playing games against

C. Both a & b

D. None of these

Correct option is C

197.Learning symbolic representations of concepts known as:

A. Artificial Intelligence

B. Machine Learning

C. Both a & b

D. None of these

Correct option is A

198.The field of study that gives computers the capability to learn without being explicitly programmed _____

A. Machine Learning

B. Artificial Intelligence

C. Deep Learning

D. Both a & b

Correct option is A

199.The autonomous acquisition of knowledge through the use of computer programs is called

-
- A. Artificial Intelligence
 - B. Machine Learning
 - C. Deep learning
 - D. All of these

Correct option is B

200.Learning that enables massive quantities of data is known as

- A. Artificial Intelligence
- B. Machine Learning
- C. Deep learning
- D. All of these